



NORTH SYDNEY BOYS HIGH SCHOOL

2022

NSBHS TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Advanced

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- In Questions in Section II, show relevant mathematical reasoning and/or calculations

**Total marks :
100**

Section I – 10 marks (pages 3 – 6)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 8 – 34)

- Attempt Questions 11 – 33
- Allow about 2 hours and 45 minutes for this section.

(For marker's use only)

SECTION	SECTION I (MC)	SECTION II (Part 1)	SECTION II (Part 2)	TOTAL
MARK	$\overline{10}$	$\overline{38}$	$\overline{52}$	$\overline{100}$

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10

1. $2 \log_{10}(x) - \log_{10}(3x)$ is equal to

A. $\log_{10}\left(\frac{x}{3}\right)$

B. $\log_{10}(x^2 - 3x)$

C. $\frac{2 \log_{10}(x)}{\log_{10}(3x)}$

D. $-\log_{10}(x)$

2. The second derivative of the function $f(x)$ is given by $f''(x) = \frac{2x}{1+x^2}$

The interval on which the graph of $f(x)$ is concave up is

A. $x < 0$

B. $x \leq 0$

C. $x > 0$

D. $x \geq 0$

3. How many terms are there in the following geometric sequence?

$$3, 6, 12, 24, \dots, 384$$

A. 7

B. 8

C. 9

D. 10

4. The solution to the inequality $6 - x - x^2 \leq 0$ is

A. $x \leq -3$ or $x \geq 2$

B. $x \leq -2$ or $x \geq 3$

C. $-3 \leq x \leq 2$

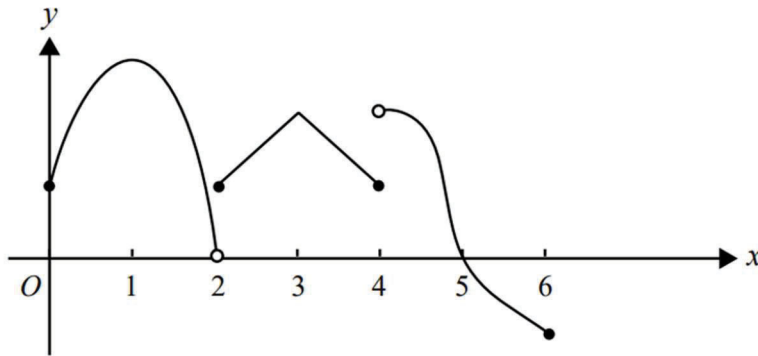
D. $-2 \leq x \leq 3$

5. If $P(A) = 0.8$, $P(B) = 0.5$ and $P(B|A) = 0.4$,

What is the value of $P(A \cap B)$?

- A. 0.32
- B. 0.25
- C. 0.1
- D. 0.5

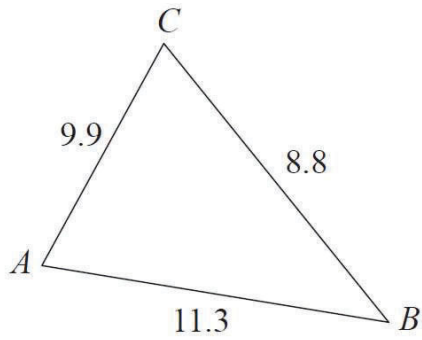
6. The graph of the function $f(x)$ with domain $x \in [0, 6]$ is shown below.



Which of the following is **not** true?

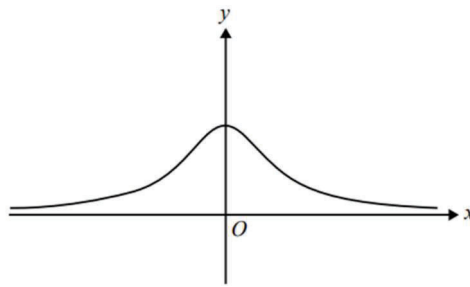
- A. The function is not continuous at $x = 2$ and $x = 4$.
 - B. The function exists for all values of x between 0 and 6.
 - C. $f(x) = 0$ for $x = 2$ and $x = 5$
 - D. The function is positive for $x \in [0, 5)$
7. Determine $\int \frac{e^x + 1}{e^x} dx$
- A. $x - e^{-x} + C$
 - B. $x + e^{-x} + C$
 - C. $1 + xe^{-x} + C$
 - D. $x + xe^{-x} + C$

8. Determine the size of angle A in the following triangle.



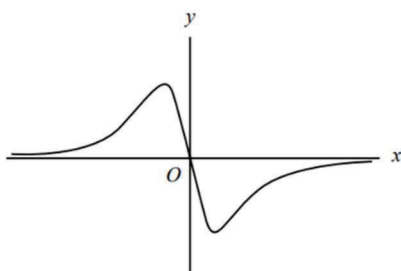
Not drawn to scale

- A. 48.5°
 B. 61.4°
 C. 118.6°
 D. 131.5°
9. The graph of a function $f(x)$ is shown below

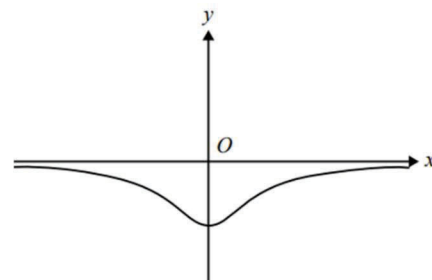


The graph of the **antiderivative** of $f(x)$ could be:

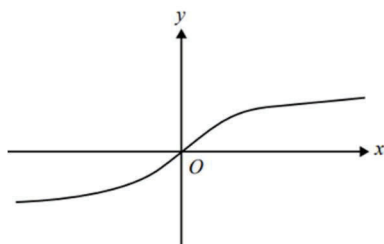
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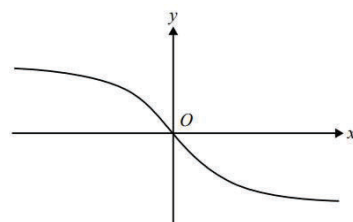
C.



B.



D.



10. If A and B are two **independent events**, then the probability of occurrence of at least one of A and B is given by:
- A. $1 + P(\bar{A})P(\bar{B})$
 - B. $1 - P(\bar{A})P(\bar{B})$
 - C. $P(A) + P(B) - P(\bar{A})P(\bar{B})$
 - D. $1 - P(A)P(B)$

Question 11 (1 mark)

Write 132° in radian measure

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Question 12 (3 marks)

A discrete random variable has the following probability distribution.

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x	1	2	3	4
$P(X = x)$	0.5	0.2	0.1	0.2

Find the **expected value** and **variance** of X

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Question 13 (6 marks)

Differentiate the following with respect to x :

(a)

$y = \sqrt{4 - x^2}$

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(b)

$y = \frac{\log_e x}{6x}$

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(c)

$y = e^{2x} \sin x$

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Question 14 (6 marks)

(a) Find $\int 6x + 7 \cos\left(\frac{x}{2}\right) dx$ **2**

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(b) Find $\int \frac{x + 2}{x^2 + 4x - 6} dx$ **2**

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(c) Evaluate $\int_0^2 3^{-x} dx$ **2**

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Question 15 (3 marks)

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Solve $\sqrt{3} \sin x = -\cos x$ for $x \in [0, 3\pi]$

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Question 16 (3 marks)

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If $\sin \theta = -\frac{12}{13}$, find the values of $\cos \theta$ and $\tan \theta$ where $\theta \in \left[\pi, \frac{3\pi}{2}\right]$

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Question 17 (2 marks)

Show that $\cot A + \frac{\sin A}{1 + \cos A} = \operatorname{cosec} A$

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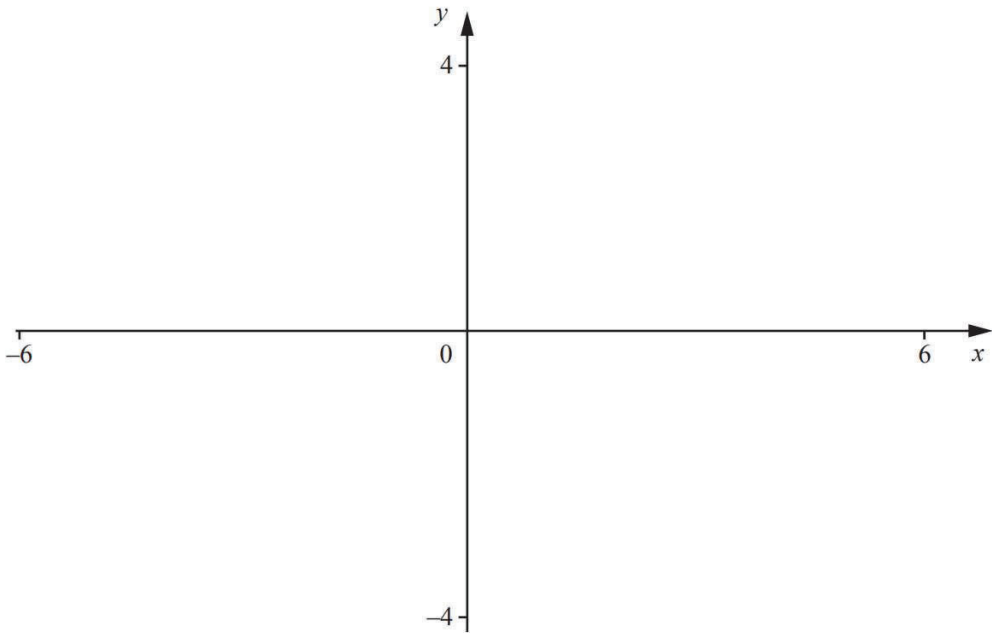
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Question 18 (5 marks)

- (a) On the axes below, sketch the graphs of $y = |x - 3|$ and $y = \left|\frac{2}{5}x\right|$, giving the coordinates of the points where the graphs meet.

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- (b) Hence, solve the inequation $\left|\frac{2}{5}x\right| < |x - 3|$

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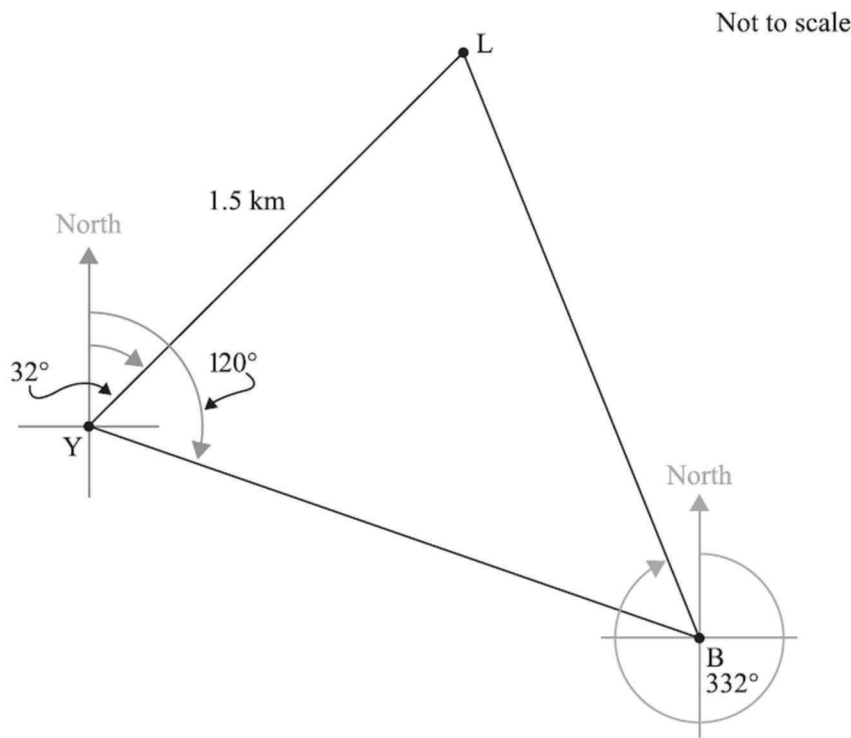
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Question 19 (6 marks)

A yacht is located at point Y and is sailing on a bearing of $032^\circ T$ towards a lighthouse at point L 1.5 km from point Y . From Y , the yacht's navigator spots a boat at point B bearing $120^\circ T$. The bearing of the lighthouse from the boat is $332^\circ T$.



- (a) Calculate the distance between the yacht and the boat.

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- (b) From the boat, the top of the lighthouse has an elevation of 2° . Determine the height of the lighthouse above sea level.

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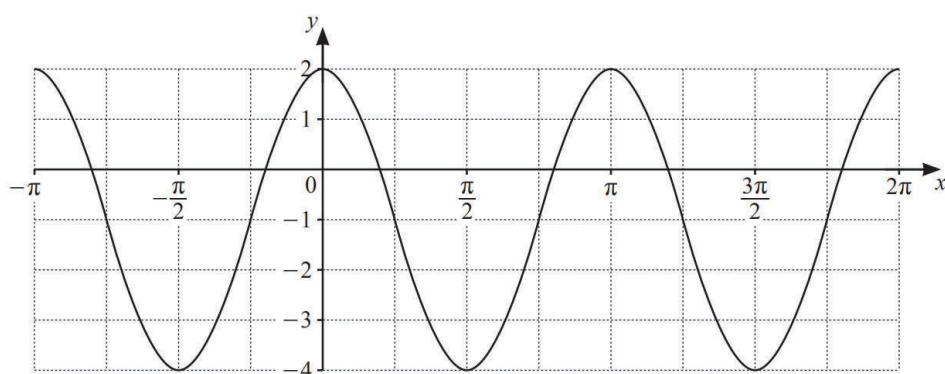
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Question 20 (3 marks)



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The curve has equation $y = a \cos bx + c$ where a, b, c are integers. Find the values of a, b and c .

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End of Questions in Answer Booklet 1

Question 21 (3 marks)

Solve the following equation for x

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$$5^x - \frac{8}{5^x} = 2$$

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Question 22 (3 marks)

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Use the trapezoidal rule with 4 intervals to evaluate $\int_{0.5}^{2.5} \sqrt[3]{\log_e x} \, dx$ correct to 3 decimal places

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Question 23 (5 marks)

- (a)

Find the x coordinate(s) of the stationary point(s) on the curve $y = 3 \log_e x + x^2 - 7x$, where $x > 0$.

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- (b)

Hence, determine the nature of each of the stationary point(s)

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Question 24 (4 marks)

There are some red counters and some white counters in a bag. At the start, 7 of the counters are red and the rest of the counters are white. Woody takes two counters from the bag. First he takes at random a counter from the bag. He does not put the counter back in the bag. Woody then takes at random another counter from the bag.

- (a)

Let the number of white counters in the bag be x .

2

Draw a tree diagram that represents the above scenario, showing all relevant information.

- (b)

It is given that the probability that the first counter Woody takes is white, **and** the second counter Woody takes is red is $\frac{21}{80}$.

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Find the number of white counters in the bag at the start.

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Question 25 (6 marks)

A geometric sequence is such that its sum of the first 4 terms is 17 times its sum of the first 2 terms. It is given that the common ratio of this geometric sequence is positive and not equal to 1.

- (a) Find the common ratio of this geometric sequence. 3

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- (b) Given that the 6th term of this geometric sequence is 64. Find the first term. 2

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- (c) Does this sequence have a limiting sum? Explain your reasoning. 1

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Question 26 (3 marks)

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Determine the number of solutions for $\sin \theta = -0.7$, where $\theta \in [0, 51\pi]$ and provide reasons for your answer.

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Question 27 (3 marks)

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The Richter scale defines the magnitude of an earthquake as $M = \log_{10} \left(\frac{I}{S} \right)$ where I is the intensity of the earthquake wave, and S is the intensity of the smallest detectable wave.

An earthquake that registered 6.4 in magnitude was followed by another which was 4 times more intense.

Determine the magnitude of the second earthquake accurate to 1 decimal place.

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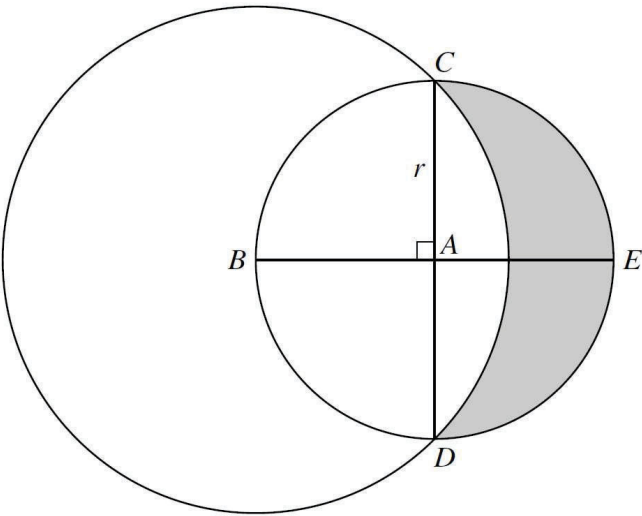
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The above diagram shows a circle with centre A and radius r . Diameters CAD and BAE are perpendicular to each other. A larger circle has centre B and passes through C and D .

Find the area of the shaded region in terms of r .

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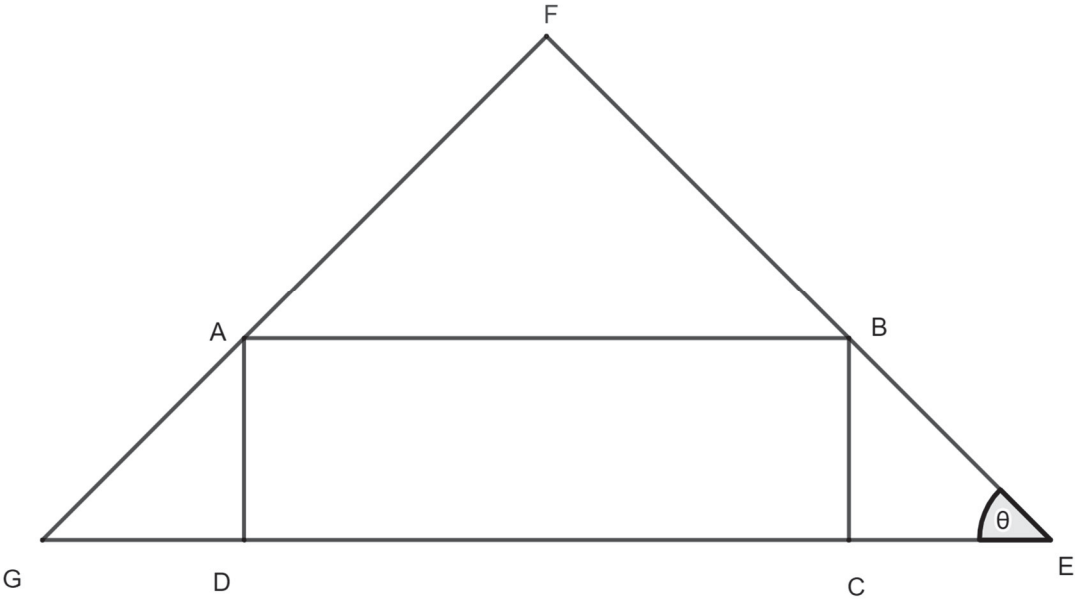
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Question 29 (7 marks)



A rectangle $ABCD$ of length 6cm and width 2cm is inscribed in an isosceles triangle EFG , where $FG = FE$.

Let $\angle FEG = \theta$

- (a) Show that the area of the isosceles triangle can be expressed as

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$$A = 12 + 9 \tan \theta + 4 \cot \theta$$

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Question 30 (4 marks)

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The first four numbers of an arithmetic sequence are $p, 9, 3p - q, 3p + q$.

Find the 2022th term.

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Question 31 (6 marks)

(a) Given that $y = xe^{-x}$, find $\frac{dy}{dx}$ and hence show that

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$$\int xe^{-x} dx = -xe^{-x} - e^{-x} + C$$

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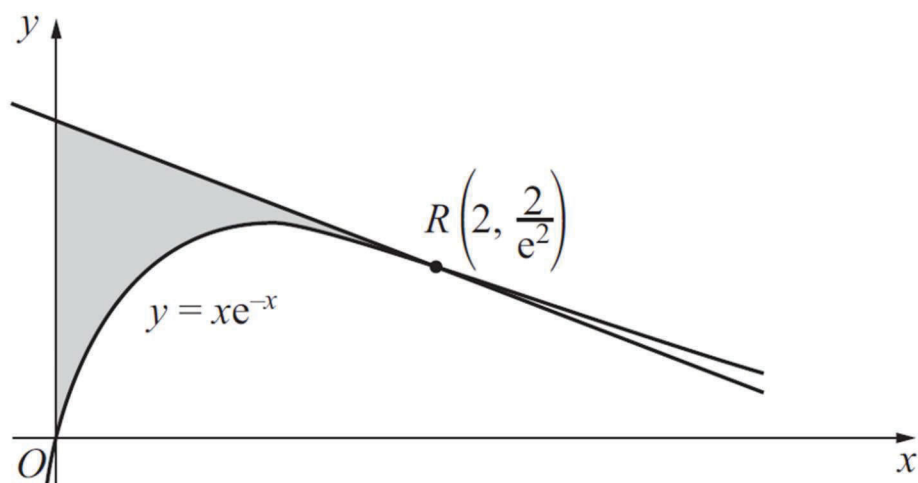
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Find the area of the shaded region bounded by the curve, the tangent and the y axis. Leaving your answer in exact form.

[illegible]

Question 32 (2marks)

Given that $E(aX + b) = aE(X) + b$, where $E(X)$ is the expected value of a discrete random variable X and a and b are constants.

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Prove that $Var(aX + b) = a^2 Var(X)$

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Question 33 (3 marks)

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At the point $(2, -5)$ on the curve $y = f(x)$, the tangent has the equation $2x - y - 9 = 0$.
Determine the equation of the tangent to the curve $y = 4 - 2f\left(3 + \frac{x}{2}\right)$ at the point $(-2, 14)$,
showing all reasoning.

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End of Paper

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10

1. $2 \log_{10}(x) - \log_{10}(3x)$ is equal to

A. $\log_{10}\left(\frac{x}{3}\right)$

B. $\log_{10}(x^2 - 3x)$

C. $\frac{2 \log_{10}(x)}{\log_{10}(3x)}$

D. $-\log_{10}(x)$

$$\begin{aligned} & \log_{10}(x^2) - \log_{10}(3x) \\ &= \log_{10}\left(\frac{x^2}{3x}\right) \\ &= \log_{10}\left(\frac{x}{3}\right) \end{aligned}$$

(A)

2. The second derivative of the function $f(x)$ is given by $f''(x) = \frac{2x}{1+x^2}$

The interval on which the graph of $f(x)$ is concave up is

A. $x < 0$

B. $x \leq 0$

C. $x > 0$

D. $x \geq 0$

concave up

when $f''(x) > 0$

$$\begin{aligned} \therefore 2x &> 0 \\ x &> 0 \end{aligned}$$

(C)

3. How many terms are there in the following geometric sequence?

3, 6, 12, 24, ..., 384

A. 7

B. 8

C. 9

D. 10

$$T_n = ar^{n-1}$$

where $a = 3$ $r = 2$

$$384 = 3 \times 2^{n-1}$$

$$128 = 2^{n-1}$$

$$2^7 = 2^{n-1}$$

$$\therefore n = 8$$

(B)

4. The solution to the inequality $6 - x - x^2 \leq 0$ is

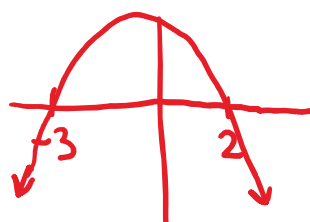
A. $x \leq -3$ or $x \geq 2$

B. $x \leq -2$ or $x \geq 3$

C. $-3 \leq x \leq 2$

D. $-2 \leq x \leq 3$

$$(3+x)(2-x) \leq 0$$



$\therefore$

$$x \leq -3 \text{ or } x \geq 2$$

(A)

5. If $P(A) = 0.8$, $P(B) = 0.5$ and $P(B|A) = 0.4$,

What is the value of $P(A \cap B)$?

- A. 0.32
B. 0.25
C. 0.1
D. 0.5

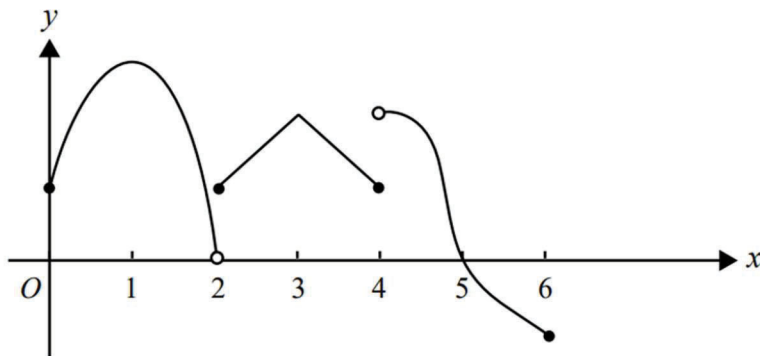
$$P(B|A) = \frac{P(B \cap A)}{P(A)}$$

$$0.4 = \frac{P(B \cap A)}{0.8}$$

$$P(B \cap A) = 0.4 \times 0.8 = 0.32$$

(A)

6. The graph of the function $f(x)$ with domain $x \in [0, 6]$ is shown below.



Which of the following is **not** true?

- A. The function is not continuous at $x = 2$ and $x = 4$. True
B. The function exists for all values of x between 0 and 6. True
C. $f(x) = 0$ for $x = 2$ and $x = 5$ False, at $x=2$ $y > 0$
D. The function is positive for $x \in [0, 5)$

(C)

7. Determine $\int \frac{e^x + 1}{e^x} dx$

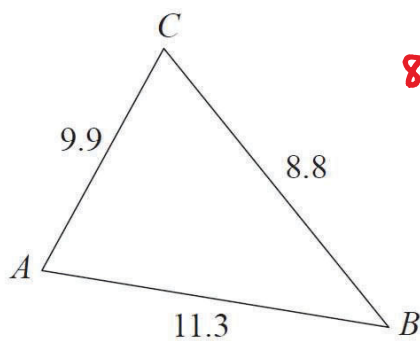
- A. $x - e^{-x} + C$
B. $x + e^{-x} + C$
C. $1 + xe^{-x} + C$
D. $x + xe^{-x} + C$

$$\int 1 + e^{-x} dx$$

$$= x - e^{-x} + C$$

(A)

8. Determine the size of angle A in the following triangle.



$$8.8^2 = 9.9^2 + 11.3^2 - 2 \times 9.9 \times 11.3 \cos A$$

$$\cos A = \frac{9.9^2 + 11.3^2 - 8.8^2}{2 \times 9.9 \times 11.3}$$

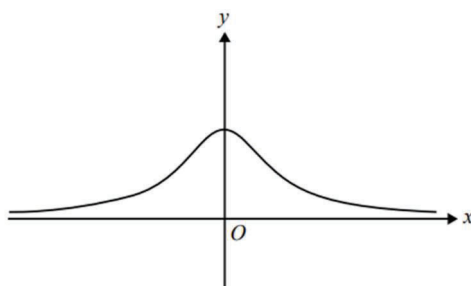
$$A \doteq 48.5^\circ$$

Not drawn to scale

(A)

- A. 48.5°
- B. 61.4°
- C. 118.6°
- D. 131.5°

9. The graph of a function $f(x)$ is shown below

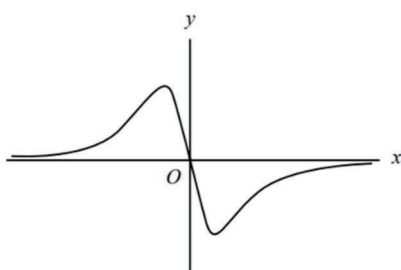


$$f'(x) > 0$$

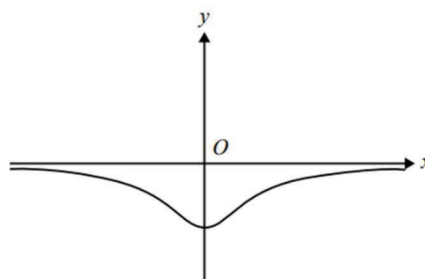
$\therefore f(x)$ is strictly increasing

The graph of the **antiderivative** of $f(x)$ could be:

A.

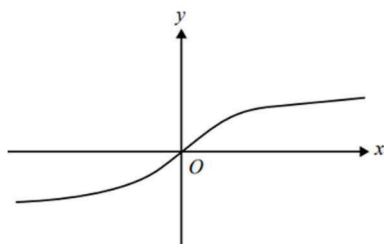


C.

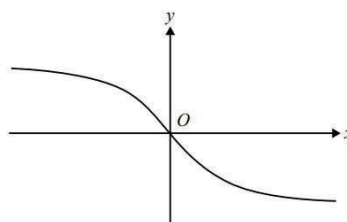


(B)

B.



D.



10. If A and B are two **independent events**, then the probability of occurrence of at least one of A and B is given by:

- A. $1 + P(\bar{A})P(\bar{B})$
- B. $1 - P(\bar{A})P(\bar{B})$
- C. $P(A) + P(B) - P(\bar{A})P(\bar{B})$
- D. $1 - P(A)P(B)$

at least one of A and B
is equivalent to $P(A \cup B)$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= P(A) + P(B) - P(A)P(B)$$

(For independent
events,

$$P(A \cap B) = P(A) \cdot P(B)$$

$$= P(A) + P(B)(1 - P(A))$$

$$= P(A) + P(B) \cdot P(\bar{A})$$

$$= 1 - P(\bar{A}) + P(B) \cdot P(\bar{A})$$

$$= 1 - P(\bar{A})[1 - P(B)]$$

$$= 1 - P(\bar{A})P(\bar{B})$$

(B)

Question 11 (1 mark)Write 132° in radian measure

1

$$132^\circ = \frac{132}{180} \pi$$

$$= \frac{11}{15} \pi$$

Question 12 (3 marks)

A discrete random variable has the following probability distribution.

3

x	1	2	3	4
$P(X = x)$	0.5	0.2	0.1	0.2

Find the **expected value** and **variance** of X

x	1	2	3	4	Sum
$x P(X=x)$	0.5	0.4	0.3	0.8	2
$x^2 P(X=x)$	0.5	0.8	0.9	3.2	5.4

$$E(x) = \sum x P(X=x)$$

$$= 2$$

$$\text{Var}(x) = E(x^2) - [E(x)]^2$$

$$= 5.4 - 2^2$$

$$= 1.4$$

Question 13 (6 marks)

Differentiate the following with respect to x :

(a) $y = \sqrt{4 - x^2}$ 2

$$y = (4 - x^2)^{\frac{1}{2}}$$
$$\frac{dy}{dx} = \frac{1}{2} (4 - x^2)^{-\frac{1}{2}} \cdot -2x$$
$$= -\frac{x}{\sqrt{4 - x^2}}$$

(b) $y = \frac{\log_e x}{6x}$ 2

$$\frac{dy}{dx} = \frac{6x \cdot \frac{1}{x} - \log_e x \cdot 6}{(6x)^2}$$
$$= \frac{6 - 6 \log_e x}{36x^2}$$
$$= \frac{1 - \log_e x}{6x^2}$$

(c) $y = e^{2x} \sin x$ 2

$$\frac{dy}{dx} = e^{2x} \cdot \cos x + \sin x \cdot 2e^{2x}$$
$$= e^{2x} (\cos x + 2 \sin x)$$

Question 14 (6 marks)

(a) Find $\int 6x + 7 \cos\left(\frac{x}{2}\right) dx$ 2

$$= 3x^2 + 14 \sin\left(\frac{x}{2}\right) + C$$

(b) Find $\int \frac{x+2}{x^2+4x-6} dx$ 2

$$\frac{1}{2} \int \frac{2x+4}{x^2+4x-6} dx$$

$$= \frac{1}{2} \log_e |x^2+4x-6| + C$$

(c) Find $\int_0^2 3^{-x} dx$ 2

$$= -\int_0^2 3^{-x} dx$$

$$= \left[\ln\left(\frac{1}{3}\right) \cdot \left(\frac{1}{3}\right)^x \right]_0^2$$

$$= \frac{1}{\log_e 3} \cdot (3^{-2} - 1)$$

$$= \frac{1}{\log_e 3} \cdot \left(1 - \frac{1}{9}\right)$$

$$= \frac{8}{9 \log_e 3}$$

Question 15 (3 marks)

3

Solve $\sqrt{3} \sin x = -\cos x$ for $x \in [0, 3\pi]$

$$\frac{\sin x}{\cos x} = -\frac{1}{\sqrt{3}}$$

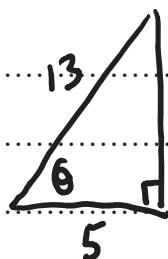
$$\tan x = -\frac{1}{\sqrt{3}}$$

$$x = \pi - \frac{\pi}{6}, 2\pi - \frac{\pi}{6}, 3\pi - \frac{\pi}{6}$$

$$= \frac{5\pi}{6}, \frac{11\pi}{6}, \frac{17\pi}{6}$$

Question 16 (3 marks)

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If $\sin(\theta) = -\frac{12}{13}$, find the values of $\cos(\theta)$ and $\tan(\theta)$ where $\theta \in \left[\pi, \frac{3\pi}{2}\right]$  θ is in quadrant 3 $\therefore \cos \theta < 0$ and $\tan \theta > 0$

$$\cos \theta = -\frac{5}{13}$$

$$\tan \theta = \frac{12}{5}$$

Question 17 (2 marks)

Show that $\cot A + \frac{\sin A}{1 + \cos A} = \operatorname{cosec} A$

2

$$\text{LHS} = \cot A + \frac{\sin A}{1 + \cos A}$$

$$= \frac{\cos A}{\sin A} + \frac{\sin A}{1 + \cos A}$$

$$= \frac{\cos A + \cos^2 A + \sin^2 A}{\sin A (1 + \cos A)}$$

$$= \frac{(\cancel{\cos A} + 1)}{\sin A (1 + \cancel{\cos A})}$$

$$= \frac{1}{\sin A}$$

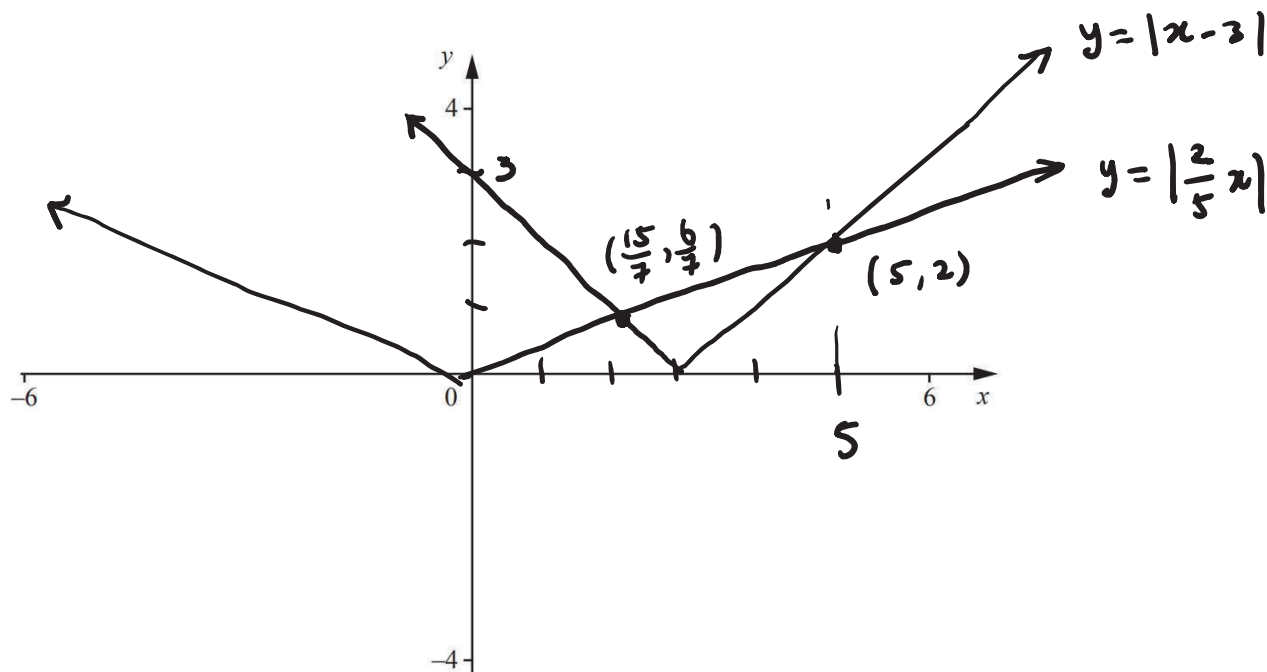
$$= \operatorname{cosec} A$$

$$= \text{RHS}$$

Question 18 (5 marks)

- (a) On the axes below, sketch the graphs of $y = |x - 3|$ and $y = \left|\frac{2}{5}x\right|$, giving the coordinates of the points where the graphs meet.

4



$$x - 3 = \frac{2}{5}x$$

$$\frac{3}{5}x = 3$$

$$x = 5$$

$$x - 3 = -\frac{2}{5}x$$

$$\frac{7x}{5} = 3$$

$$7x = 15$$

$$x = \frac{15}{7}$$

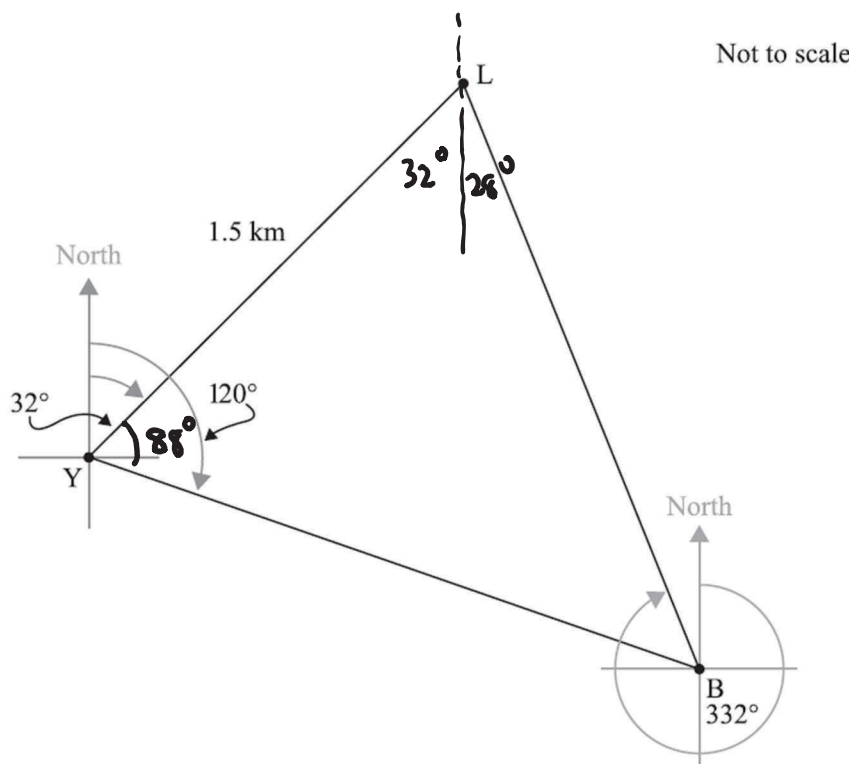
- (b) Hence, solve the inequation $\left|\frac{2}{5}x\right| < |x - 3|$

1

$$x < \frac{15}{7} \quad \text{or} \quad x > 5$$

Question 19 (6 marks)

A yacht is located at point Y and is sailing on a bearing of $032^\circ T$ towards a lighthouse at point L 1.5 km from point Y . From Y , the yacht's navigator spots a boat at point B bearing $120^\circ T$. The bearing of the lighthouse from the boat is $332^\circ T$.



- (a) Calculate the distance between the yacht and the boat.

3

$$\angle YLB = 60^\circ \quad \text{and} \quad \angle LBY = 32^\circ$$

$$\frac{YB}{\sin 60^\circ} = \frac{1.5}{\sin 32^\circ}$$

$$YB = \frac{1.5 \times \sin 60^\circ}{\sin 32^\circ}$$

$$= 2.45 \text{ km}$$

- (b) From the boat, the top of the lighthouse has an elevation of 2° . Determine the height of the lighthouse above sea level.

3

Find LB using sine rule

$$\frac{LB}{\sin 88^\circ} = \frac{1.5}{\sin 32^\circ}$$

$$\therefore LB = \frac{1.5 \sin 88^\circ}{\sin 32^\circ}$$

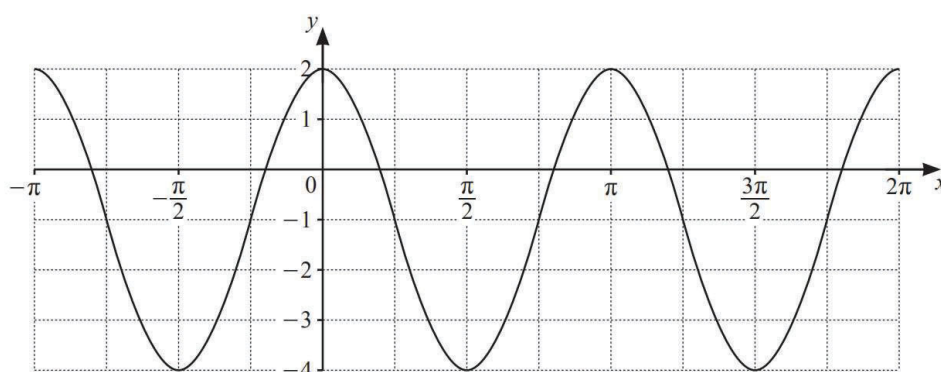
$$= 2.83 \text{ km}$$

$$\therefore \tan 2^\circ = \frac{h}{2.83}$$

$$\therefore h = 2.83 \times \tan 2^\circ$$

$$= 99 \text{ m (to nearest m)}$$

Question 20 (3 marks)



3

The curve has equation $y = a \cos bx + c$ where a, b, c are integers. Find the values of a, b and c .

Period: $T = \frac{2\pi}{b}$ amplitude: $\frac{2 - (-4)}{2}$ centre of oscillation: $\frac{2 + (-4)}{2}$

$$\therefore \pi = \frac{2\pi}{b}$$

$$= 3$$

$$= -1$$

$b = 2$ $a = 3$ $c = -1$

Question 21 (3 marks)

Solve the following equation for x

3

$$5^x - \frac{8}{5^x} = 2$$

$$\text{let } m = 5^x$$

$$\therefore x = \frac{\ln 4}{\ln 5}$$

$$m - \frac{8}{m} = 2$$

$$\div 0.86$$

$$m^2 - 8 = 2m$$

$$m^2 - 2m - 8 = 0$$

$$(m-4)(m+2) = 0$$

$$\therefore m = 4 \text{ or } m = -2$$

$$\therefore 5^x = 4 \text{ or } 5^x = -2$$

$$x = \log_5 4 \quad \text{no solutions}$$

$$= \frac{\ln 4}{\ln 5}$$

Question 22 (3 marks)

3

Use the trapezoidal rule with 4 intervals to evaluate $\int_{0.5}^{2.5} \sqrt[3]{\log_e x} dx$ correct to 3 decimal places

4 intervals $\rightarrow$ 5 function values

x	0.5	1.0	1.5	2.0	2.5
y	-0.885	0	0.74	0.885	0.97

$$\therefore \int_{0.5}^{2.5} \sqrt[3]{\log_e x} dx \div \frac{0.5}{2} [(-0.885 + 0.97) + 2(0 + 0.74 + 0.885)]$$

$$\div 0.834$$

Question 23 (5 marks)

- (a) Find the x coordinate of the stationary point(s) on the curve $y = 3 \log_e x + x^2 - 7x$, where $x > 0$.

2

$$\frac{dy}{dx} = \frac{3}{x} + 2x - 7$$

Stationary point when $\frac{dy}{dx} = 0$

$$\frac{3}{x} + 2x - 7 = 0$$

$$2x^2 - 7x + 3 = 0$$

$$(2x - 1)(x - 3) = 0$$

$$\therefore x = \frac{1}{2} \text{ or } x = 3$$

- (b) Hence, determine the nature of each of these stationary point(s)

3

$$\frac{d^2y}{dx^2} = -\frac{3}{x^2} + 2$$

$$\text{at } x = \frac{1}{2}, \frac{d^2y}{dx^2} = -3 \times 4 + 2 = -10 < 0$$

$$\therefore \text{maximum turning point at } x = \frac{1}{2}$$

$$\begin{aligned} \text{at } x = 3, \frac{d^2y}{dx^2} &= -\frac{3}{9} + 2 \\ &= \frac{5}{3} \\ &> 0 \end{aligned}$$

$$\therefore \text{minimum turning point at } x = 3$$

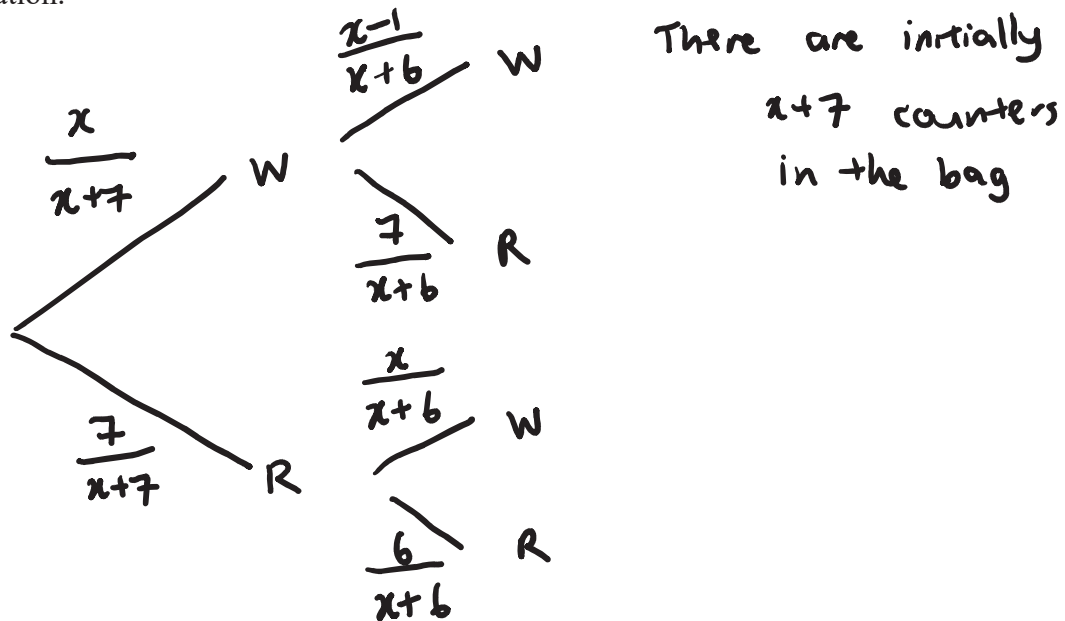
Question 24 (4 marks)

There are some red counters and some white counters in a bag. At the start, 7 of the counters are red and the rest of the counters are white. Woody takes two counters from the bag. First he takes at random a counter from the bag. He does not put the counter back in the bag. Woody then takes at random another counter from the bag.

- (a) Let the number of white counters in the bag be x .

2

Draw a tree diagram that represents the above scenario, showing all relevant information.



- (b) It is given that the probability that the first counter Woody takes is white, **and** the second counter Woody takes is red is $\frac{21}{80}$.

2

Find the number of white counters in the bag at the start.

Given $P(WR) = \frac{21}{80}$

$$\frac{x}{x+7} \times \frac{7}{x+6} = \frac{21}{80}$$

$$80x = 3(x+7)(x+6)$$

$$= 3(x^2 + 13x + 42)$$

$$= 3x^2 + 39x + 126$$

$$3x^2 - 41x + 126 = 0$$

$$x = \frac{41 \pm \sqrt{169}}{6}$$

$$= \frac{41+13}{6} \text{ or } \frac{41-13}{6}$$

$$= 9 \text{ or } 4\frac{2}{3}$$

$\therefore x = 9$ as x is an integer

Question 25 (6 marks)

A geometric sequence is such that its sum to 4 terms is 17 times its sum to 2 terms. It is given that the common ratio of this geometric sequence is positive and not equal to 1.

- (a) Find the common ratio of this geometric sequence.

3

$$S_4 = 17S_2$$

$$S_4 = \frac{a(r^4 - 1)}{r - 1} \quad \text{and} \quad S_2 = a + ar = a(1 + r)$$

$$\therefore \frac{a(r^4 - 1)}{r - 1} = 17a(1 + r)$$

$$r^4 - 1 = 17(r^2 - 1) \quad \text{as } a \neq 0$$

$$(r^2 - 1)(r^2 + 1) = 17(r^2 - 1)$$

$$(r^2 - 1)(r^2 + 1) - 17(r^2 - 1) = 0$$

$$(r^2 - 1)[r^2 + 1 - 17] = 0$$

$$\therefore (r^2 - 1)(r^2 - 16) = 0$$

$$\therefore r = 4 \quad \text{as } r \neq 1, -1 \text{ or } -4$$

- (b) Given that the 6th term of this geometric sequence is 64. Find the first term.

2

$$T_6 = ar^5$$

$$\therefore 64 = a(4)^5$$

$$a = \frac{64}{4^5}$$

$$= \frac{1}{16}$$

- (c) Does this sequence have a limiting sum? Explain your reasoning.

1

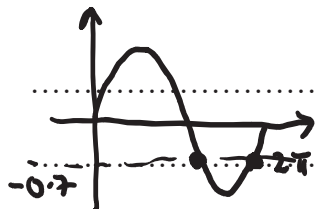
For a limiting sum to exist, $-1 < r < 1$.

as $r = 4$, this sequence will not have a limiting sum.

Question 26 (3 marks)

3

Determine the number of solutions for $\sin \theta = -0.7$, where $\theta \in [0, 51\pi]$ and provide reasons for your answer.



$\sin \theta = -0.7$ twice in each period

period = 2π

$\therefore$ for $\theta \in [0, 50\pi]$ we have 25 periods

$\therefore$ 50 solutions.

For $\theta \in [50\pi, 51\pi]$ no solutions exist
as $\sin \theta > 0$ in this interval.

$\therefore$ total of 50 solutions.

Question 27 (3 marks)

3

The Richter scale defines the magnitude of an earthquake as $M = \log_{10} \left(\frac{I}{S} \right)$ where I is the intensity of the earthquake wave, and S is the intensity of the smallest detectable wave.

An earthquake that registered 6.4 in magnitude was followed by another which was 4 times more intense.

Determine the magnitude of the second earthquake accurate to 1 decimal place.

1st Quake:

2nd Quake

$$6.4 = \log_{10} \left(\frac{I_1}{S} \right)$$

$$I_2 = 4I_1$$

$$\frac{I_1}{S} = 10^{6.4}$$

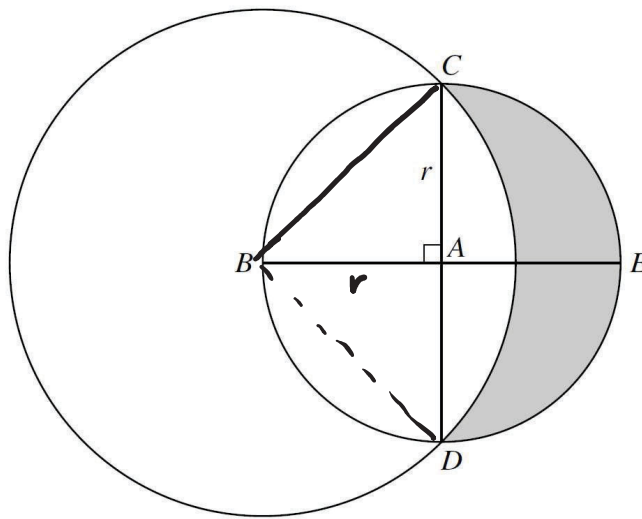
$$\therefore M_2 = \log_{10} \left(\frac{I_2}{S} \right)$$

$$= \log_{10} \left(\frac{4I_1}{S} \right)$$

$$= \log_{10} (4 \times 10^{6.4})$$

$$= \log_{10} 4 + 6.4$$

$$\doteq 7.0 \text{ (to 1 d.p.)}$$



The above diagram shows a circle with centre A and radius r . Diameters CAD and BAE are perpendicular to each other. A larger circle has centre B and passes through C and D .

Find the area of the shaded region in terms of r .

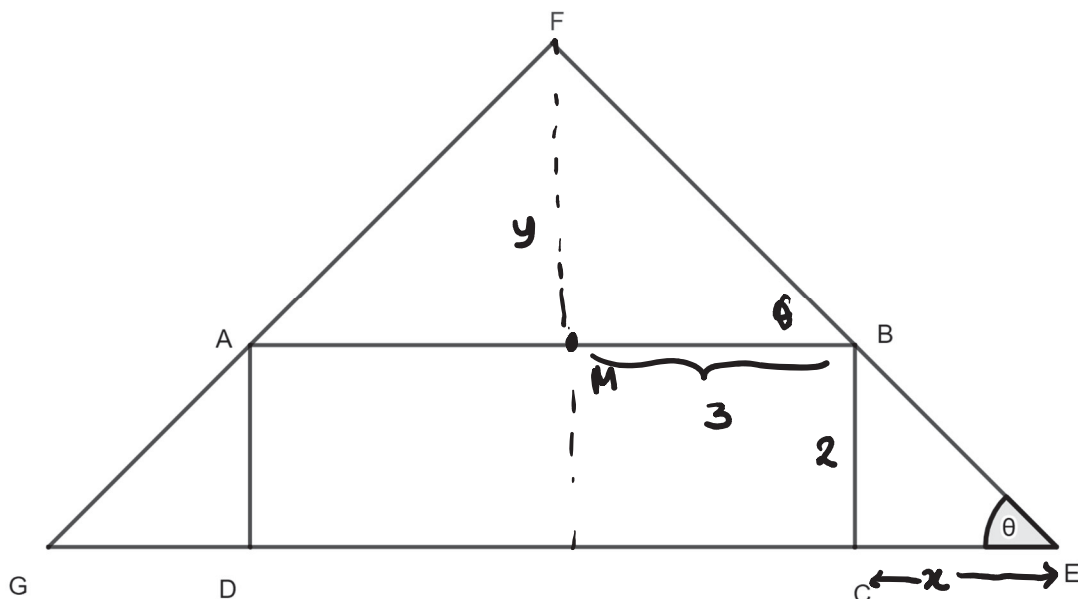
Using Pythagoras' Theorem in $\triangle ABC$, $BC = \sqrt{2}r$ and $\angle BAC = \frac{\pi}{4}$, hence $\angle BAD = \frac{\pi}{2}$

$$A_{\text{segment } CD} = \frac{1}{2} (\sqrt{2}r)^2 \left(\frac{\pi}{2} - \sin \frac{\pi}{2} \right)$$

$$= r^2 \left(\frac{\pi}{2} - 1 \right)$$

$$\begin{aligned} \text{Shaded Area} &= \text{Area of semicircle} - \text{Area of segment} \\ &= \frac{1}{2} \pi r^2 - \left(\frac{\pi r^2}{2} - r^2 \right) \\ &= \frac{1}{2} \pi r^2 - \frac{\pi r^2}{2} + r^2 \\ &= r^2 \text{ units}^2 \end{aligned}$$

Question 29 (7 marks)



A rectangle $ABCD$ of length 6cm and width 2cm is inscribed in an isosceles triangle EFG , where $FG = FE$.

Let $\angle FEG = \theta$

- (a) Show that the area of the isosceles triangle can be expressed as

3

$$A = 12 + 9 \tan \theta + 4 \cot \theta$$

$$A = \frac{1}{2} b \times h$$

$$b = 2x + 6 \quad \text{and} \quad h = y + 2$$

$$\tan \theta = \frac{2}{x}$$

$$\tan \theta = \frac{y}{3}$$

$$\therefore x = 2 \cot \theta$$

$$\therefore y = 3 \tan \theta$$

$$\therefore A = \frac{1}{2} (2x + 6)(y + 2)$$

$$= (x + 3)(y + 2)$$

$$= xy + 2x + 3y + 6$$

$$= 6 + 2(2 \cot \theta) + 3(3 \tan \theta) + 6$$

$$= 12 + 4 \cot \theta + 9 \tan \theta$$

(b) Hence, find the minimum area of the isosceles triangle that inscribes the rectangle.

4

$$A = 12 + 4(\tan \theta)^{-1} + 9 \tan \theta$$

$$\frac{dA}{d\theta} = -4(\tan \theta)^{-2} \cdot \sec^2 \theta + 9 \sec^2 \theta$$

$$= \frac{-4 \cos^2 \theta}{\sin^2 \theta} \times \frac{1}{\cos^2 \theta} + \frac{9}{\cos^2 \theta}$$

$$= \frac{-4}{\sin^2 \theta} + \frac{9}{\cos^2 \theta}$$

$$= \frac{-4 \cos^2 \theta + 9 \sin^2 \theta}{\sin^2 \theta \cos^2 \theta}$$

stationary when $\frac{dA}{d\theta} = 0$

$$-4 \cos^2 \theta + 9 \sin^2 \theta = 0$$

$$9 \sin^2 \theta = 4 \cos^2 \theta$$

$$\tan^2 \theta = \frac{4}{9}$$

$$\tan \theta = \pm \frac{2}{3} \quad \text{as } 0 < \theta < 90^\circ$$

$$\theta \doteq 33^\circ 41'$$

Testing for minimum

θ	30°	$33^\circ 41'$	45°
$\frac{dA}{d\theta}$	-4	0	10

$\therefore$ minimum when $\theta \doteq 33^\circ 41'$

$$\therefore A = 12 + 4 \cdot \frac{3}{2} + 9 \cdot \frac{2}{3}$$

$$= 24 \text{ cm}^2$$

Question 30 (4 marks)

4

The first four numbers of an arithmetic sequence are $p, 9, 3p - q, 3p + q$.

Find the 2022th term.

$$9 - p = (3p + q) - (3p - q)$$

$$= 2q$$

$$p + 2q = 9 \quad \textcircled{1} \quad \text{cm}^2$$

Now $9 + 2q = 3p - q$ as $2q$ is common difference.

$$3p - 3q = 9$$

$$p - q = 3 \quad \textcircled{2}$$

$$\cdot \quad \textcircled{1} - \textcircled{2}$$

$$3q = 6$$

$$q = 2$$

$$\therefore p = 5$$

$\therefore$ sequence is $5, 9, 13, 17, \dots$

$$a = 5, d = 4$$

$$\therefore T_{2022} = 5 + (2021) \times 4$$

$$= 8089$$

Question 31 (6 marks)

- (a) Given that $y = xe^{-x}$, find $\frac{dy}{dx}$ and hence show that

3

$$\int xe^{-x} dx = -xe^{-x} - e^{-x} + C$$

$$\begin{aligned}\frac{dy}{dx} &= x \cdot -e^{-x} + e^{-x} \\ &= -xe^{-x} + e^{-x}\end{aligned}$$

$$\therefore \int -xe^{-x} + e^{-x} dx = xe^{-x} + C,$$

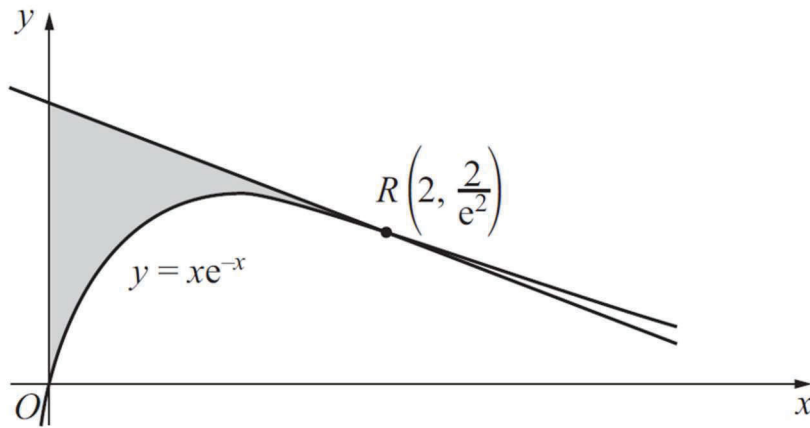
$$- \int xe^{-x} dx = xe^{-x} - \int e^{-x} dx$$

$$\int xe^{-x} dx = -xe^{-x} + \int e^{-x} dx$$

$$= -xe^{-x} - e^{-x} + C$$

(b)

3



The diagram above shows part of the curve $y = xe^{-x}$ and the tangent to the curve at the point $R(2, \frac{2}{e^2})$.

Find the area of the shaded region bounded by the curve, the tangent and the y axis.

From a) $\frac{dy}{dx} = e^{-x}(1-x)$

when $x=2$ $m_T = -\frac{1}{e^2}$

$\therefore$ equation of tangent: $y - \frac{2}{e^2} = -\frac{1}{e^2}(x-2)$

$$y = -\frac{1}{e^2}x + \frac{2}{e^2}$$

$$y = -\frac{1}{e^2}x + \frac{4}{e^2}$$

Area: $\int_0^2 -\frac{1}{e^2}x + \frac{4}{e^2} - xe^{-x} dx$

$$= \left[-\frac{1}{2e^2}x^2 + \frac{4}{e^2}x - (-xe^{-x} - e^{-x}) \right]_0^2$$

$$= \left[-\frac{1}{2e^2}x^2 + \frac{4}{e^2}x + xe^{-x} + e^{-x} \right]_0^2$$

$$= \left(-\frac{2}{e^2} + \frac{8}{e^2} + \frac{2}{e^2} + \frac{1}{e^2} \right) - (0 + 0 + 0 + 1)$$

$$= \frac{9}{e^2} - 1$$

Question 32 (2marks)

Given that $E(aX + b) = aE(X) + b$, where $E(X)$ is the expected value of a discrete random variable X and a and b are constants.

2

Prove that $\text{Var}(aX + b) = a^2 \text{Var}(X)$

$$\begin{aligned}\text{Var}(aX+b) &= E((aX+b)^2) - [E(aX+b)]^2 \\ &= E(a^2X^2 + 2abX + b^2) - [aE(X) + b]^2 \\ &= a^2E(X^2) + 2abE(X) + b^2 - a^2(E(X))^2 - 2abE(X) - b^2 \\ &= a^2(E(X^2) - (E(X))^2) \\ &= a^2 \text{Var}(X)\end{aligned}$$

Question 33 (3 marks)

3

At the point $(2, -5)$ on the curve $y = f(x)$, the tangent has the equation $2x - y - 9 = 0$. Determine the equation of the tangent to the curve $y = 4 - 2f\left(3 + \frac{x}{2}\right)$ at the point $(-2, 14)$, showing all reasoning.

Tangent will be transformed in the same way as $f(x)$

$$y = 4 - 2f\left(3 + \frac{x}{2}\right)$$

$$\Rightarrow -\left(\frac{y-4}{2}\right) = f\left(3 + \frac{x}{2}\right)$$

replace y with $-\left(\frac{y-4}{2}\right)$ and x with $\left(3 + \frac{x}{2}\right)$

$$2\left(3 + \frac{x}{2}\right) - \left(\frac{4-y}{2}\right) - 9 = 0$$

$$12 + 2x + y - 4 - 18 = 0$$

$$y = -2x + 10$$

End of Paper

Section II Extra writing space

If you use this space, clearly indicate which question you are answering.

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